

MELINA BAGHER

Tehran-Iran

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🌐 Personal Website

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🌐 LinkedIn page

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EDUCATION

🎓 MASTER OF SCIENCE IN BIOMEDICAL ENGINEERING

Sep. 2019 – Sep. 2022

Sharif University of Technology

Tehran, Iran

- **Thesis:** “Comparative Analysis of Haplotype Assembly Algorithms to Identify Optimal Methods”
- **Advisor:** Prof. Babak Hosseini Khalaj, Prof. Mehran Jahed
- **GPA:** 19.19/20 → 4/4

🎓 BACHELOR OF SCIENCE IN BIOMEDICAL ENGINEERING

Sep. 2014 – Sep. 2018

Islamic Azad University, Science and Research Branch

Tehran, Iran

- **Thesis:** “Extraction of Fractal Dimension of Malignant and Benign Skin Lesions in Dermoscopic Images”
- **Advisor:** Prof. S. Rashidi
- **GPA:** 19.10/20 → 4/4

PUBLICATIONS

📖 JOURNAL PAPERS

○ Bagher M., Karimzadeh R., khalaj B.H., Jahed M; *A rapid heuristic algorithm to solve the single individual Haplotype assembly problem* ; AUT Journal of Electrical Engineering 55 (2), 191-206, 2023.

📖 CONFERENCE PAPERS

○ Bagher M., Karimzadeh R., khalaj B.H., Jahed M; *QuickHap: a Quick heuristic algorithm for the single individual Haplotype reconstruction problem* ; 29th National and 7th International Iranian conference on Biomedical Engineering 2022 (ICBME).

○ Karimzadeh R., Bagher M., Khodabakhsh A., Arabi H., Zaidi H; *Generative Adversarial-based Framework for Classification using Imbalance Data: Application to Pneumonia Detection in Chest Radiographs*; IEEE NSS and Medical Imaging Conference 2022.

SELECTED ACADEMIC PROJECTS

📖 Artificial Intelligence and Deep Learning Projects/Tasks

☆ GAN-Based Framework for Pneumonia Detection

Developed a GAN-based classification framework in PyTorch using feature-space augmentation to address class imbalance in chest X-rays. (*Published in IEEE NSS/MIC*)

☆ Bone fracture classification using multi-level feature fusion

Developed a transfer learning model by fusing intermediate features from InceptionV3 into a custom classification head, improving performance through multi-scale feature representation. (*Self-initiated project*)

☆ CycleGAN for Image-to-Image Translation

Built a CycleGAN in PyTorch for unsupervised image-to-image translation using dual generator–discriminator architecture and cycle-consistency loss. (*Self-initiated project to explore generative adversarial models*)

☆ Enhancing Adversarial Robustness in Convolutional Neural Networks

Implemented FGSM and PGD attacks in PyTorch and trained a CNN to improve robustness while maintaining clean accuracy. (*Course project on adversarial machine learning*)

☆ Image generation using VAE and Latent Diffusion pipeline

Implemented a latent diffusion pipeline (Stable Diffusion components) including U-Net denoising, VAE decoding, and text conditioning for text-to-image generation. (*Self-initiated project on diffusion-based generative models*)

☆ License Plate Recognition using CNN-based Sequence Modeling

Developed a CNN-based model for Iranian license plate recognition, performing fixed-length character-wise classification directly from input images using supervised learning. (*Self-initiated project*)

📁 Medical Image Processing Projects/Tasks

☆ Spinal Cord CT Registration Pipeline

Developed a registration pipeline for spinal cord CT images using CPD and ICP algorithms and evaluated performance using HD, Dice Score, and ASD metrics. (*Course project in Medical Image Processing and Analysis*)

☆ Dermoscopy Image Preprocessing and Segmentation

Applied filtering and morphological image processing techniques in MATLAB for artifact removal and skin lesion segmentation. (*In line with my BSc project*)

📁 Computational Genomics and Bioinformatics Projects/Tasks

☆ CNN for Single Individual Haplotype Reconstruction

Designed a single-layer CNN in PyTorch for haplotype assembly and developed a custom cost function to improve reconstruction accuracy. Evaluated performance using MEC score. (*Project related to my M.Sc. thesis*)

☆ NGS Data Simulation and Variant Calling Pipeline

Simulated sequencing data using PBsim and NanoSim and implemented a full variant calling pipeline using BWA-MEM, GATK, Samtools, and BCFtools. (*Aligned with my M.Sc. thesis*)

HONORS & AWARDS

- Ranked 1st among approximately 500 participants in the nationwide Ph.D. entrance exam for Biomedical Engineering (Bioelectric), 2023
- Ranked 2nd among approximately 900 participants in the nationwide Ph.D. entrance exam for Biomedical Engineering, 2026
- Ranked 39th among approximately 40,000 participants in the nationwide M.Sc. entrance exam for Electrical Engineering
- Member of National Elite Foundation
- Distinguished B.Sc. graduate

TEACHING EXPERIENCE

- **TEACHING ASSISTANT** | Medical Images Analysis and Processing | Supervisor: Prof. Emad Fatemizadeh
Department of Electrical Engineering, Sharif University of Technology *Fall 2021*
- **TEACHING ASSISTANT** | System Biology | Supervisor: Prof. Babak Hossein Khalaj
Department of Electrical Engineering, Sharif University of Technology *spring 2020*
- **TEACHING ASSISTANT** | Electronics 2 | Supervisor: Prof. Keyvan Maghouli
Department of Biomedical Engineering, Islamic Azad University, Science and Research Branch *Fall 2019*
- **TUTOR** | Engineering Mathematics, Differential Equations, Signals and Systems,
Mahan Institute of Higher Education *Summer 2019*

SELECTED Extracurricular COURSES

☆ Supervised Machine Learning; Regression and Classification

Andrew NG, DeepLearning.AI, Stanford University via Coursera, Jan. 2024
Instructor:Andrew Ng

☆ Advanced Learning Algorithms

DeepLearning.AI, Stanford University via Coursera, Jan. 2024
Instructor:Andrew Ng

☆ Introduction to Machine Learning and DeepLearning

Rayan.AI, Sharif Univercity of Technology, Sep. 2024
Instructor:Mohammad Hossein Rohban, Rayan.AI Group

☆ Trustworthiness in DeepLearning

Rayan.AI, Sharif Univercity of Technology, Sep. 2024
Instructor:Mohammad Hossein Rohban, Rayan.AI Group

TECHNICAL SKILLS

Programming Languages:	Python, R, C/C++, Bash
Python Selected Libraries:	Pytorch, TensorFlow, Scikit-learn, Matplotlib, Numpy, Scipy
Version Control:	Git, GitHub
Software Simulators:	Matlab, Simulink
Hardware Simulator:	Pspice, Proteus
Typesetting:	T _E X, Prezi, Microsoft Office(Word, Powerpoint, Excel)
Operating Systems:	Windows,Ubuntu
NGS Tools and Simulators:	GATK Toolkit, BWA, Samtools, BCFtools, PBSIM, NanoSim, ART

RESEARCH INTERESTS

- Medical Image Processing, Computer Vision
- Medical Signal Processing
- Machine learning, Deep Learning
- Computational Genomics, NGS data analysis

LANGUAGES

English TOEFL-iBT: 98 (R:28, L:21, S:23, W:26)

French A1

Persian/Farsi Native

HOBBIES

SPORTS: Volleyball, Badminton, Table Tennis

MUSIC: Piano (Advanced), Guitar (Intermediate)

WORKING EXPERIENCE

Medical Equipment unit of Parsian Hospital

Summer 2018

ENGINEERING INTERNSHIP

- Becoming familiar with Repair and maintenance of health monitoring equipment.

Educational counsellor

2019 - 2021

Mahan Institute of Higher Education

- Consulting to Master and Ph.D Konkoor Candidates.

Private Piano Tutor

2016-Present

Teaching piano to students of various age groups and skill levels

REFERENCES

Prof. Babak Hosseini Khalaj, Department of Electrical Engineering, Sharif University of Technology

Email: khalaj@sharif.edu

Asst. Prof. Mehran Jahed, Department of Electrical Engineering, Sharif University of Technology

Email: Jahed@sharif.edu

Asst. Prof. Saeed. Rashidi, Department of Biomedical Engineering, Islamic Azad University, Science and Research Branch

Email: rashidi.saeid@iau.ac.ir